

PEERACHET KHANITSON

AI/MACHINE LEARNING ENGINEER • DATA SCIENCE • DATA ENGINEER

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Portfolio: <https://peerachet-khanitson.vercel.app>

SUMMARY

Computer Engineering graduate with a strong focus on Artificial Intelligence and Machine Learning. I have hands-on experience analyzing complex datasets and building predictive models for real-world scenarios. I am highly motivated to transform data into actionable insights, and eager to apply my technical knowledge to develop intelligent solutions while growing as an AI Engineer.

EDUCATION

Mae Fah Luang University - Bachelor of Engineering - Computer Engineering | GPAX: 3.68

WORK EXPERIENCE

Mitsubishi Electric Thai Auto-Parts Co., Ltd. Internship

Jan - Apr 2026 (4 Months)

- Data Collection & System Development: Digitized operational workflows using a centralized Power Apps solution, ensuring standardized and structured data collection for reliable downstream analysis.
- Data Pipeline & Automation: Built automated data syncing workflows via Power Automate, significantly improving data integrity by eliminating manual data entry errors.
- Data Structuring & UI Design: Engineered structured data schemas and designed intuitive UIs to boost operational efficiency and facilitate accurate data capture from end-users.

PROJECTS

Senior Project (Machine Learning model training) <https://github.com/peerachetkns/PM2.5>

- Developed an end-to-end multi-class ML pipeline to classify PM2.5 levels according to the Thai Air Quality Standard.
- Processed and engineered massive meteorological datasets using techniques like missing value imputation and trigonometric wind encoding.
- Addressed severe class imbalance by systematically evaluating multiple oversampling algorithms (e.g., SMOTE variants).
- Evaluated 8 ML/DL architectures, with the Extra Trees and Borderline-SMOTE model yielding an outstanding 92.7% Accuracy, 92.7% F1-Score, and 99.3% ROC-AUC.

Credit Scoring Model (Logistic Regression) <https://github.com/peerachetkns/Credit-Scoring-Model>

- Developed a Logistic Regression credit scoring model to assess loan default risk.
- Optimized features using WOE, IV, and DSR, and resolved multicollinearity to validate accuracy via ROC-AUC and Gini metrics.
- Created a Score Grading framework with operational cut-off simulations to optimize loan approval ratios and reduce bad debt.

Real-Time Object Detection <https://github.com/peerachetkns/Real-Time-Object-Detection-project>

- Constructed a real-time video object detection system using OpenCV's DNN module and a pre-trained MobileNet SSD.
- Classified 20 object categories, utilizing a 0.3 confidence threshold to filter predictions for high accuracy.
- Implemented dynamic bounding boxes and text overlays to visualize class labels and confidence scores.

Haar Cascade Image Detection <https://github.com/peerachetkns/Face-Detection>

- Developed an OpenCV script using Haar Cascade classifiers to detect human faces, eyes, and cats in static images.
- Optimized detection via grayscale conversion and ROI extraction, visualizing results with dynamic bounding boxes.

TECHNICAL SKILLS

- Programming Languages: Python, C, Java
 - ML, DL & NLP: Classification, Regression, Clustering, Ensemble (RF, ET, GB), Deep Learning (CNN, LSTM), Natural Language Processing (NLP), Predictive Analytics
 - CV & Image Processing: OpenCV, Object Detection, Face Detection, Image Enhancement, Spatial/Frequency Filtering, Restoration, Color Processing
 - Model Evaluation: Cross-Validation, Bias/Variance, Key Metrics (Accuracy, Precision, F1, ROC-AUC)
 - Data Analysis & Prep: EDA, Data Cleaning, Feature Engineering, Transformation, Imbalanced Handling
 - Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, TensorFlow/Keras, OpenCV
 - Tools & Platforms: VS Code, Jupyter Notebook, Google Colab, Git & GitHub, Microsoft Power Platform (Power Apps, Power Automate)
 - Web and Mobile App Development: HTML, CSS, React, JavaScript, Flutter (Dart)
 - Databases: MySQL, Firebase Firestore
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CERTIFICATE

- Machine Learning & Data Analytics: Predicting Consumer Behavior for Business Decisions
<https://app.futureskill.co/api/certificate?courseId=121&userId=325172>
 - Natural Language Processing (NLP): Discovering Data Insights & Solving Business Problems with NLP and Deep Learning
<https://app.futureskill.co/api/certificate?courseId=114&userId=325172>
 - Computer Vision & OpenCV: Building Professional CV Systems, Basic Object/Face Detection, and Deep Neural Network Workshop
<https://app.futureskill.co/api/certificate?courseId=271&userId=325172>
 - Artificial Intelligence: In-Depth Neural Networks & AI
<https://app.futureskill.co/api/certificate?courseId=67&userId=325172>
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SOFT SKILLS

- Collaboration – Worked collaboratively with peers on a Machine Learning project.
- Adaptability & Active Learning – Proactive in acquiring new knowledge and skills independently.
- Problem Solving – Skilled at debugging and resolving real-world code issues in project work.
- Communication – Able to communicate effectively in both spoken and written English.